



PLC Based Automatic Door Opening & Closing System

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Abstract

The PLC (Programmable Logic Controller) Based Automatic Door Opening & Closing System is a modern solution designed to enhance convenience, efficiency, and safety in various environments such as commercial buildings, residential complexes, hospitals, and industrial facilities. This system utilizes advanced PLC technology to automate the operation of doors, offering seamless functionality and reliable performance.

The primary objective of this system is to automate the process of opening and closing doors in response to user input or environmental conditions. By employing sensors and actuators, the system detects the presence of individuals approaching the door and initiates the appropriate action, whether it be opening or closing the door. This functionality improves accessibility and convenience for users while also optimizing energy efficiency by minimizing the duration of open-door states.

Key components of the PLC Based Automatic Door Opening & Closing System include sensors for detecting motion, presence, and obstacles, actuators for controlling the movement of doors, and a PLC unit responsible for processing inputs, executing control logic, and coordinating the operation of the system. The PLC unit is programmed with specific algorithms to ensure smooth and accurate door operation while also incorporating safety features to prevent accidents or collisions.

Moreover, the system offers flexibility in configuration, allowing for customization according to the specific requirements of different environments. It can be integrated with existing building management systems, access control systems, and security systems to create a comprehensive solution that enhances overall functionality and security.

In conclusion, the PLC Based Automatic Door Opening & Closing System represents a significant advancement in door automation technology, offering benefits such as improved accessibility, efficiency, and safety. Its versatility and reliability make it an ideal choice for various applications, contributing to enhanced user experience and operational efficiency in diverse settings.

1. Introduction

The figure of the automatic door in life can be seen everywhere. The main working principle of the automatic door is to sense whether there is an object passing through the sensing device at the top of the door and the auxiliary infrared sensing device on the side of the door frame, giving the door a switch signal and then programmable control. The device, the frequency converter, the drive motor and the transmission control the movement of the door leaf to realize the function of opening and closing the door. The biggest feature of the automatic door is that it can complete the automatic door opening and closing action without being supervised, which gives people great convenience, but the biggest problem in the case of no one is the safety hazard. The popularity of automatic doors in China is not wide enough, which

leads to the slow development of automatic doors. The system construction relies mainly on imports. The control part of the hardware cannot meet the market demand. The software design is relatively simple, and the designed system is relatively simple. These can determine the automatic door, the stability and safety of operation, so China needs to strengthen the research of

control system, combined with the foreign automatic door function and local actual needs, using advanced software compilation, solving various problems existing in the development process of automatic door, and promoting the automatic door industry. The design of this paper is an automatic door applied to public places such as shopping malls and high-end hotel entrances and exits, which require strong security and practicality. Integrating automatic door scenes and usage habits in daily life, this paper designs the most used automatic sliding door. The automatic sliding door uses the inductive switch and safety grating to realize automatic control. At the same time, the manual control is realized by using the operation buttons, and the use is made. Auxiliary devices such as backup power supply to meet the needs of different situations of adult traffic.

1.1 System Design

The design of the automatic door needs to be considered for the purpose of safety and simplicity. This design is applied to public places with large traffic flow. It is designed to take care of the practicality of practical use and improve convenience. The design of automatic door control system based on PLC needs to consider the practicality and safety in real life, and ensure the safety of the passing object under the guarantee of smooth operation of the automatic door. This paper adopts the photoelectric sensor to collect whether the object passes the signal to the PLC, and controls the opening and closing of the automatic door through the inverter, the driving motor and the transmission device in the PLC control actuator; meanwhile, the infrared sensor is opened after the automatic door is opened. Let the relevant implementing agencies coordinate work to ensure that the automatic door can operate normally and safely, as shown in Figure 1.1. The main content of the automatic door control system design process includes the following aspects: master the control principle and control flow of the automatic door, and understand the control process of the automatic door as a whole. The input signal device and the output control device are clearly designed, and the number of input and output points is estimated in combination with the requirements of the automatic door function design. Select the appropriate PLC type based on the number of I/O points.

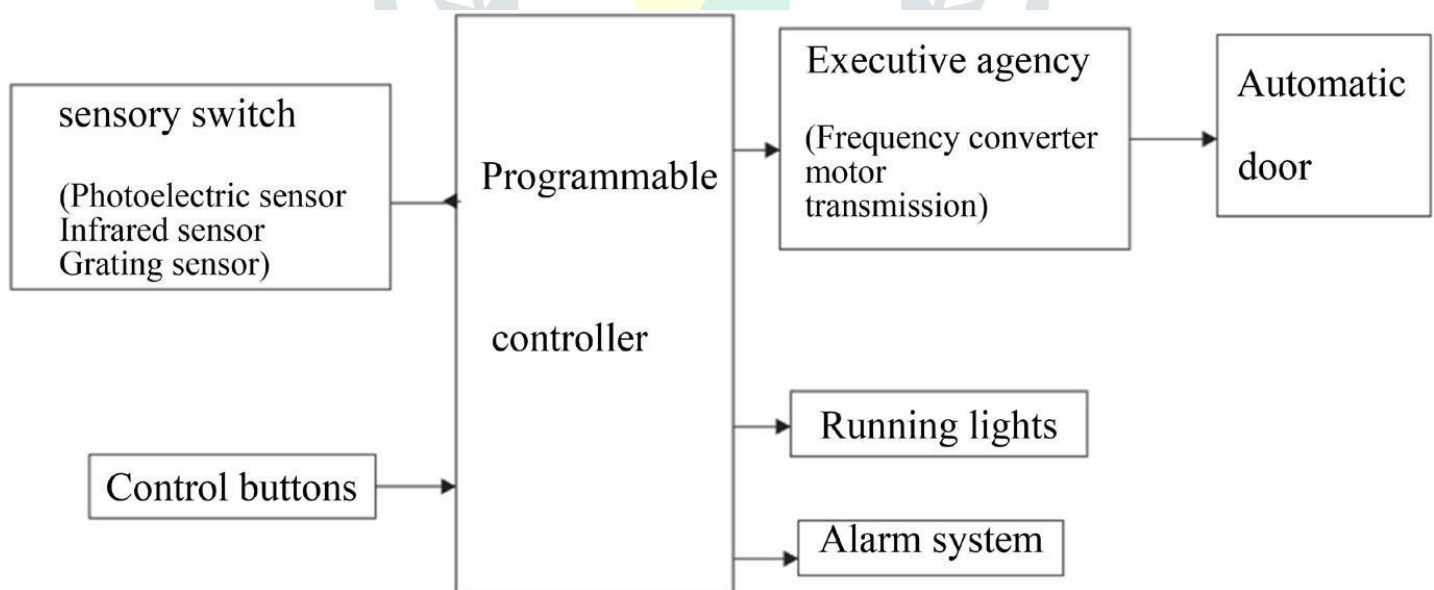


Figure 1.1 Schematic diagram of the automatic door control system.

According to the number of I/O points, the input and output addresses are reasonably distributed, and the terminal wiring diagram is drawn, which is beneficial to the design of the ladder diagram and the convenience in using the simulation. Draw a complete ladder diagram according to the curriculum design requirements and practical applications. Use software simulation to export the program to the specified format and run the simulation to see if the function is implemented. Troubleshoot and unreasonable troubleshooting procedures, optimize program design; optimize overall debugging of automatic door systems, optimize problematic programs, and continuously improve the system through repeated simulations until the desired functions are implemented.

1.2 System Hardware Design

In this paper, the drive motor drives the transmission device to realize the opening and closing of the automatic door. The operation of the motor is realized by the AC frequency controlled by the PLC to control the frequency converter to supply the motor. The advantages of this design are easy operation, stable operation and high real-time performance. Compared with other types of motors, more mature AC motors can control cost, have stable torque output and a larger speed range. Improve the stability of the long-term operation of the automatic door, the system hardware components, as shown in Figure 1.2. The transmission of the automatic door is mainly composed of a drive motor, a transmission belt and a transmission pulley. Considering the convenient installation and maintenance, the transmission only occupies a small part of the whole system. For example, the pulley that drives the door body needs to be fixed outside the wall to be detachable structure. The three-phase asynchronous motor is selected as the drive motor. Passing three-phase symmetrical alternating current into the stator winding of the motor generates a circular rotating magnetic field. In the magnetic field, the rotor winding cuts the magnetic field to generate an induced current, and the rotor winding in the closed path also generates an electromagnetic force. Under the interaction of the two magnetic fields, the motor, the electromagnetic torque in the same direction as the rotating magnetic field is formed on the rotating shaft, so that the rotor of the motor rotates.

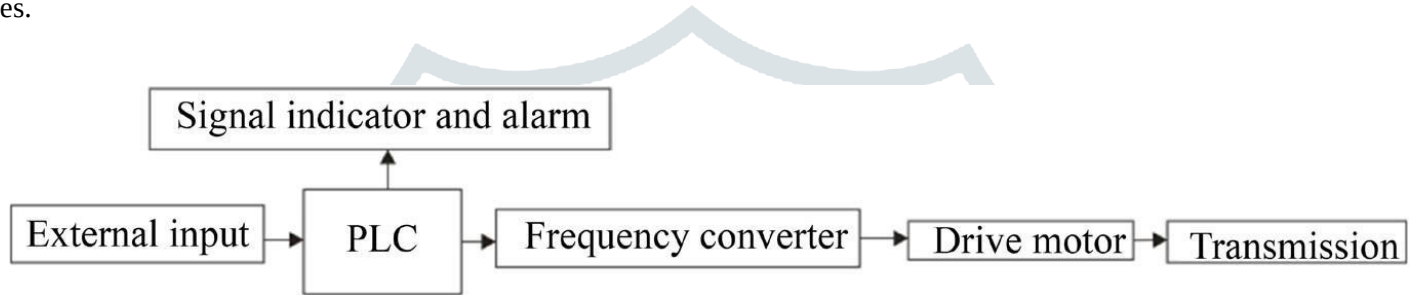


Figure 1.2 Hardware structure of the system.

1.3 System Software Design.

The basic functions realized by this design are as follows:

- 1) When there is no data connection with the PLC, the new file can be edited according to actual needs, the error location is checked, the program and system configuration are modified, and the program is stored in the hard disk of the PC.
- 2) Under the online communication, the software uploads and downloads data and programs, and can directly operate the PLC, such as editing programs and modifying instructions. When writing a program is an error, the software can point out some of the errors entered. When writing a ladder diagram, when there is a program problem, the dialog window below can directly indicate the problem of the network program that has the problem, and the user can modify the program according to the prompt.
- 3) At the menu bar of the software, the user can use the response function module according to the prompt.
- 4) The user can modify the PLC configuration, communication settings, and work type, and can also monitor the operation of the PLC. The system flow is shown in Figure 1.3. Automatic control of closing and anti-clamping: The automatic door touches the door opener I0.4 and enters the 8s timing. During the three sensors I0.6, I0.7, I1.0, if an object is detected, the timer T37 retimed. After the automatic door detects no object for 8 s, the PLC's Q0.1, M0, V0 action, pass to the inverter a motor reversal signal and low speed running signal, a total of 6s automatic door closing, if any object is closed when the automatic door is closed I1.0 detects that the automatic door stops closing and executes the automatic door opening procedure. In the automatic closing, due to the fact that the object passes the fast factor, the automatic door uses a safety grating with a delay of only 0.2s to prevent the object from being clamped.

There are two kinds of alarm conditions in the automatic door. The first one is the automatic door program. The automatic door appears to be operated manually or automatically (Q0.0 or Q0.2) and closed (Q0.1 or Q0.3). The door will judge by the program, the M0.1 action will stop the whole automatic door control system, the fault light Q1.1 will light up, remind the staff to come to check, after checking no problem, press the abnormal reset button I1.3 to make the automatic the gate program is reset.

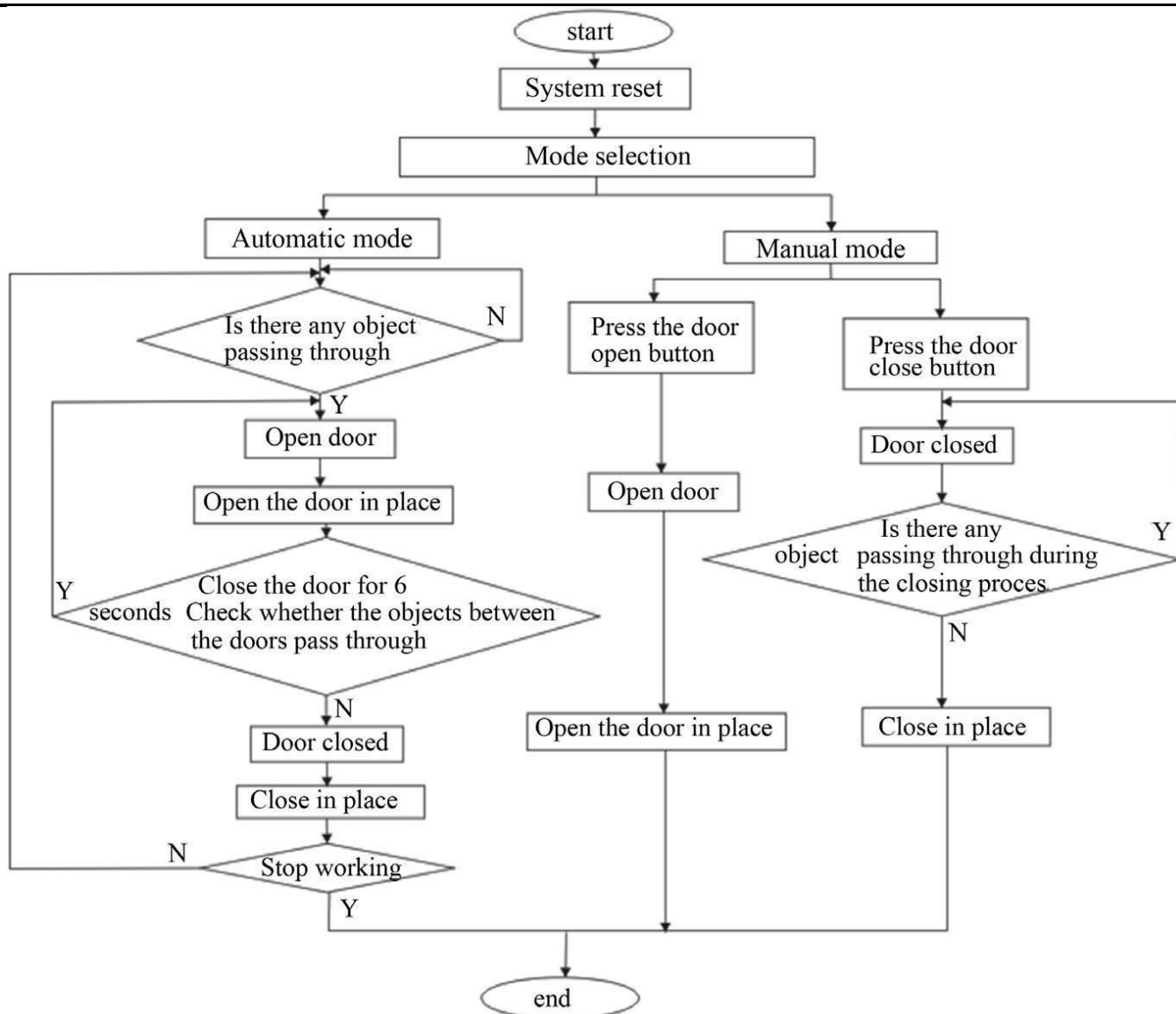


Figure 1.3. System flow chart.

When a person or an object is not detected by the automatic door, in order to prevent the secondary control from being caused by the manual control, the surrounding personnel can directly press the red emergency stop button I0.1, and the M0.0 action automatic door program stops working, and then for rescue or treatment, the alarm light Q0.5 works simultaneously with the alarm Q0.4 to remind the problem.

1.4 System Verification.

When the automatic door encounters an emergency, and presses the emergency stop button I0.1, the automatic door stops all work, the alarm Q0.4 and the alarm light Q0.5 work at the same time. After the check is normal, it presses the abnormal reset button I1.3 automatically. The door is re-worked. The automatic door in the text is mainly composed of two control modes, manual control and automatic control of the opening and closing of the automatic door. When the automatic control is run, the actuator is opened and closed according to whether or not someone has changed the frequency of the inverter through the system.

1.5adder Diagram

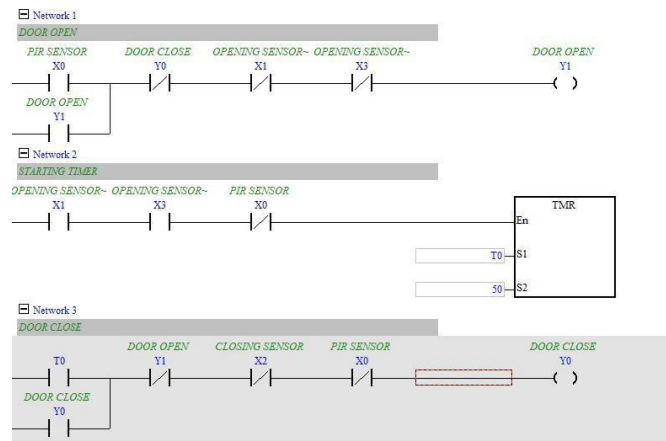


Figure 1.5 (i) Ladder Diagram which is made on ISPSoft Programming Software

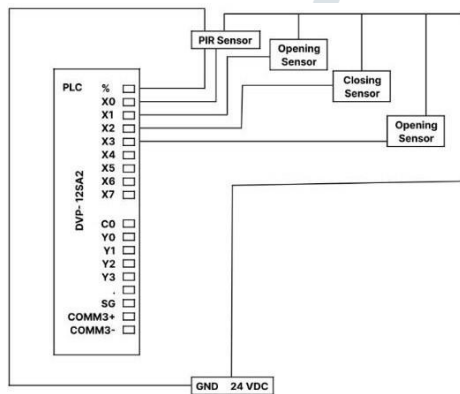


Figure 1.5 (ii) PIR Sensor to PLC (Input)

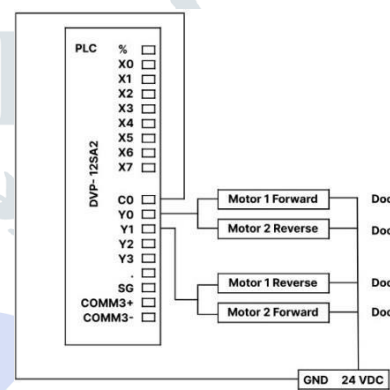


Figure 1.5 (iii) PLC to Motor (Output)

2. Literature Review

PLC-based door opening and closing systems have gained significant attention in various industrial and commercial applications due to their efficiency, reliability, and automation capabilities. This literature review explores the existing knowledge and research on this technology.

The development of PLC technology marked a significant shift in the control and automation of various processes, including door operations. Early PLCs were used for simple tasks, but over time, they evolved into versatile controllers for various applications.

PLCs are widely employed in door control systems due to their ability to execute complex logic, integrate with various sensors, and provide real-time control. These controllers have proven to be a fundamental component in achieving precise and efficient door automation.

2.1 Advantages of PLC-Based Systems

Numerous studies highlight the benefits of PLC-based door control systems. These advantages include improved safety, energy efficiency, reduced human intervention, and the ability to integrate with other building management systems”^[1]

2.2 Applications of PLC Door Systems

Research demonstrates the broad range of applications for PLC-based door systems. These systems are used in manufacturing plants for material handling, in warehouses for improved logistics, and in commercial spaces for enhancing security and accessibility”^[2]

2.3 Sensor Technologies

Sensors are pivotal in PLC-based door systems. Researchers have explored the use of different sensor technologies, such as proximity sensors, motion detectors, and RFID systems. These sensors detect the presence of objects or individuals, enabling the PLC to respond accordingly” [3]

2.4 Actuation Mechanisms

Actuators play a crucial role in the door control process. Studies have investigated the use of various actuators, including electric motors, solenoids, and pneumatic cylinders, to physically move doors in response to PLC commands” [4]

2.5 PLC Programming for Door Control

Research has delved into the programming aspects of PLCs in door control. Researchers have developed complex control algorithms that consider multiple inputs from sensors, user preferences, and safety protocols to automate door movements effectively” [5]

2.6 Safety and Security Features

Safety and security are paramount in door control systems. Research has examined the integration of emergency stop functions, obstacle detection, and access control to enhance the safety of PLC-based door systems” [6]

2.7 Challenges and Limitations

Several studies acknowledge the challenges associated with PLC-based door systems, such as maintenance requirements, system complexity, and potential security vulnerabilities. Researchers have also explored solutions to mitigate these challenges” [7]

2.8 Integration with Building Management Systems

The integration of PLC-based door systems with broader building management systems is an active area of research. Studies have explored ways to synchronize door operations with heating, ventilation, and air conditioning systems, lighting, and security systems for better energy efficiency and coordination” [8]

2.9 Future Trends and Research Directions

The literature indicates several emerging trends, such as the adoption of IoT technology for remote monitoring and control, machine learning for predictive maintenance, and energy optimization through data analytics. These trends reflect the ongoing evolution of PLC-based door systems” [9]

PLC-based door opening and closing systems have become a crucial part of industrial and commercial automation. The research reviewed here highlights the technology's advantages, applications, and ongoing developments, which contribute to increased efficiency, safety, and convenience in various settings. Future research is likely to focus on the integration of advanced technologies to further enhance the capabilities of these systems.

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3. Problem Statement.

The current manual door access systems utilized in various environments such as commercial buildings, educational institutions, and residential complexes often suffer from inefficiencies, security vulnerabilities, and lack of user convenience. Traditional systems reliant on physical keys or swipe cards are susceptible to unauthorized access, cumbersome administration of credentials, and limited scalability. Moreover, the absence of automated mechanisms results in reduced operational efficiency and increased reliance on human intervention for access control.

To address these challenges, there is a need for an advanced automatic door system integrated with RFID authentication technology. This system should leverage the capabilities of Programmable Logic Controllers (PLCs) and Arduino microcontrollers to automate door operations while ensuring secure and convenient access for authorized users.

The primary problems to be addressed by this project include:

3.1 Security Vulnerabilities:

Current manual door access systems may be susceptible to unauthorized entry due to lost or stolen keys/cards, unauthorized duplication, or compromised access codes.

3.2 Operational Inefficiencies:

Manual door operation necessitates physical effort and time for users to open/close doors, leading to delays and congestion during peak usage periods.

3.3 Lack of Scalability and Flexibility:

Existing door access systems may lack scalability to accommodate growing user populations or evolving access control requirements.

3.4 User Convenience and Accessibility:

Manual entry systems may pose challenges for users with mobility impairments or special needs, hindering their ability to independently access controlled areas.

Addressing these issues through the development and implementation of a PLC-based automatic door system integrated with Arduino-based RFID authentication technology will enhance security, efficiency, and user experience in accessing controlled environments. This project seeks to provide a comprehensive solution that mitigates security risks, streamlines access control processes, and improves accessibility for users while offering scalability and flexibility to adapt to evolving requirements.

4. Methodology

4.1 Outlining Hardware Specifications

* Outlining the hardware specifications like sensors, motors, RFID readers, and microcontrollers, considering factors such as power consumption, power output.

4.2 PLC Programming:

- * Developing a PLC logic using ladder logic to control the automatic door mechanism.
- * Utilizing ISP Soft software and a Delta communication cable to transfer the ladder logic to the PLC for sensor data processing, monitoring door states, and facilitating decision-making.

4.3 Arduino Development:

- * Programing the Arduino UNO microcontroller to interface with RFID readers for authentication purposes.
- * Creating code in the Arduino IDE software and uploading it to the Arduino UNO board to capture RFID tag data, authenticate against stored credentials, and relay authentication status to the PLC.

4.4 Wiring

Designing an input/output circuit diagram for both the PLC and Arduino systems, incorporating a PIR sensor connected to the input pins of the PLC and a motor linked to the output pins via wiring. Additionally, integrating a 24VDC SMPS to guarantee stable power supply for the PLC.

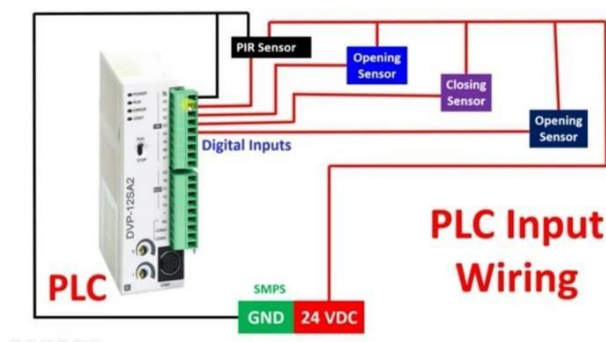


Figure 4.4 (i) PLC Input Wiring

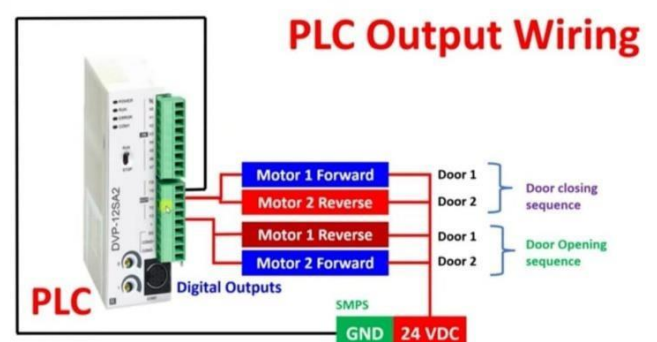


Figure 4.4 (ii) PLC Output Wiring

4.5 Integration of Components:

* Integrating the PLC and Arduino components to form a automatic door eco-system.

4.6 Testing and Validation:

* Conducting thorough testing to verify the functionality, performance, and reliability of the automatic door system integrated with RFID authentication.

5. Conclusion

The PLC-based automatic door opening and closing system lays a solid foundation for future advancements and enhancements. One promising avenue for further development is the integration of advanced sensor technologies to enhance the system's capabilities. For instance, the implementation of advanced computer vision algorithms could enable the system to detect and classify objects more accurately, thereby improving obstacle detection and avoidance.

Moreover, exploring the potential for cloud connectivity could revolutionize the system's functionality. By enabling remote monitoring and control via the internet, users could manage door operations from anywhere, enhancing convenience and accessibility. Additionally, leveraging data analytics techniques could provide valuable insights into usage patterns and system performance, facilitating predictive maintenance and optimization.

Furthermore, there is scope for expanding the application of the system to various settings, such as healthcare facilities, commercial buildings, and public spaces. Tailoring the system to meet the specific needs of different environments could unlock new opportunities for innovation and impact.

Additionally, integrating sustainability features, such as energy-efficient components and renewable energy sources, could align the system with emerging environmental standards and contribute to long-term sustainability goals.

Overall, the future scope of the project is vast and multifaceted, offering numerous possibilities for further research, development, and real-world deployment.

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